

SEEIR: Semantic driven End-to-end Exploration and Inspection for optimizing 3D Reconstruction Quality and Coverage*

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Abstract—In this paper, we introduce a Reinforcement Learning (RL) approach to a 3D inspection task on an Unmanned Ground Vehicle (UGV). We learn an end-to-end joint policy for navigation and object inspection by

Index Terms—Reinforcement Learning, 3D Gaussian Splatting, Robot Inspection and Exploration.

I. INTRODUCTION

We propose a Semantic-driven Active perception algorithm that uses Deep Reinforcement learning (DRL) methods together with classical Information gain methods for exploration to tackle robot inspection. We decouple our approach by attaching an actuateable Camera, known as a Pan-Tilt camera, to an Unmanned Ground Vehicle (UGV). We use an egocentric Occupancy map for Exploration and a Surface visibility map for coverage measurement. We train an agent to autonomously inspect the coverage of a semantically unknown target in an unstructured, clustered warehouse environment. We also develop an RL training environment for training inspection policies, which we open source, that supports training 128 environments concurrently

II. OUTLINE

The outline of this paper includes Section III, which covers the state of the art and a literature review of robot inspection methods, including classification and learning methods; and Section VII, which goes over the methodology, the reward design, and network inputs.

III. RELATED WORKS

IV. PROBLEM FORMULATION

Our objective is to autonomously navigate an unknown environment to locate a 3D object with a semantic target ξ and surface $\mathcal{S}$. We represent the surface $\mathcal{S}$ as a discrete set of observable faces $F = \{f_1, f_2, \dots, f_N\}$, where N is the total number of faces. The robot aims to maximize the visual

coverage of the set F and the inspection quality $Q(f_i) \in [0, 1]$ of each observed face f_i . We formulate the inspection task as a Partially Observable Markov Decision Process (POMDP).

- Action Space ($\mathcal{A}$): The action $a_t \in \mathbb{R}^5$ controls the base and the camera optics simultaneously:

$$a_t = [v_t, \omega_t, \dot{\theta}_{pan}, \dot{\theta}_{tilt}, \dot{f}_t]$$

Where v, ω are the differential drive linear and angular velocities, $\dot{\theta}$ represents the angular velocities of the camera joints, and $\dot{f}_t$ is the rate of change of the focal length (zoom).

- Observation Space ($\mathcal{O}_t$) We define our observation space obtained at every time step as

$$\mathcal{O}_t = [I_t, \Omega_t, \mathcal{M}_t, s_t]$$

I_t represents the RGBD Image information from the navigation and inspection cameras, Ω_t is a binary image obtained by semantic masking out our object of interest, and $\mathcal{M}_t$ encodes the environment's spatial information as maps. $s_t = [p_t, q_t, v_t, \omega_t, u_{t-1}, \theta_t^{pan}, \theta_t^{tilt}]$ represents the proprioceptive state of the robot of 3D position p_t , orientation q_t as 4D quaternions, 3D linear and angular velocity v_t, ω_t , controller actions u_{t-1} from previous time step, as well as the joint positions of the pan θ_t^{pan} and tilt articulations θ_t^{tilt} of our PTZ camera in radians.

V. ROBOT SETUP AND SENSOR SUITE

A. Navigation Camera

Our Navigation is an RGBD camera mounted on the front of our Jackal robot. This camera depth data is used to generate our occupancy map, intended for navigating the unknown environment, and the depth data is used with ray tracing to generate our occupancy map

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B. Pan-Tilt-Zoom (PTZ)/Inspection Camera

Attached to the top of the Jackal, we add an articulation to simulate the Pan and Tilt function of a PTZ Camera. The camera is an RGBD and, similar to the Navigation camera, is used to generate the Surface Visibility Map. This Camera is responsible for inspecting our inspection target ξ , and we run our semantic segmentation masking through this camera.

C. Ray Caster Camera

The Ray caster camera detects surface face IDs of our inspection target

D. IMU

VI. ROBOT INSPECTION AND EXPLORATION

A. Map Representation

To provide our agent with the spatial features of our environment, we provide our agent with a local egocentric map.

1) *Occupancy Channel* $M_{occ} \in [0, 1]$: Using the front Navigation Camera, we generate an Occupancy map, where each voxel encodes the probability of a cell being occupied. This provides the agent with spatial information for collision-free path planning to the goal. Where the value of a voxel $\sqsubseteq_i$ encodes the probability $p(v_{i,j,k})$ of a voxel being occupied

2) *Surface Visibility Channel*: $M_{vis} \in [0, 1]$: Using the Inspection camera attached to the robot's Pan-Tilt-Zoom articulation, we generate a surface visibility map via ray tracing, tracking only occupied cells. This provides the agent with a spatial representation of the current surfaces observed.

3) *Visitation Map* M_{visit} : Using the robot's odometry information, we maintain a 2D map that tracks the robot's visitation history.

B. Robot Exploration

To enable our robot to find our inspection target ξ , we use an information-theoretic object to reduce the map's entropy. To encourage generalizable exploration policies and learning the spatial representation of the environment, we aim to minimize the robot's uncertainty about the map.

Using the Shannon entropy definition, we define the entropy of a voxel $v_{i,j,k}$ as the binary entropy of its occupancy probability $p_{i,j,k} = p$:

$$\mathcal{H}(v_{i,j,k}) = -[p \log(p) + (1 - p) \log(1 - p)] \quad (1)$$

Consequently, the total entropy of the map M , assuming independence between voxels, is the summation of the entropy of all voxels:

$$\mathcal{H}(M) = \sum_{i,j,k} \mathcal{H}(v_{i,j,k}) \quad (2)$$

C. Surface Visibility Coverage

The exploration described in Section VI-B is driven by observations from the navigation camera; however, this does not guarantee that we will observe the inspection target with our PTZ Camera. For this, we use an objective function that maximizes the information gain of the Surface Visibility map. To encourage map coverage, we use a saturation function for observing a surface. The current visibility score $\mathcal{S}(\mathcal{V}_t)$ is a weighted sum over the visibility map cells $v_{i,t}$:

$$\mathcal{S}(\mathcal{V}_t) = \sum_{i=1}^N -\gamma_v e^{-k_v v_{i,t}} \quad (3)$$

D. Exploration Objective

To explore the unknown environment and locate our inspection target ξ , we aim to learn a policy π that maximizes the joint objective of increasing visibility coverage while reducing map uncertainty. We define the total objective J as:

$$J(\pi) = \mathcal{S}(\mathcal{V}_t) - \lambda \mathcal{H}(M) \quad (4)$$

Our goal is to find the optimal policy $\pi^* = \arg \max_{\pi} J(\pi)$.

E. Inspection Quality Measure

We introduce a bounded quality function for the coverage of a surface face, $q(A_f) \in [0, 1]$. To encourage detailed inspection, we define the information gain A_f as the cumulative weighted pixel count of face f . To encourage efficient coverage of all surfaces of the inspection target, we model the view quality with diminishing returns, and we adapt the submodular objectives proposed in [1], suitable for a single camera capable of zooming to optimize for A_f

$$q(A_f) = 1 - \exp\left(\frac{-A_f}{\tau}\right)$$

$$A_f = \sum_{p \in \mathcal{P}_f} (\hat{n}_f \cdot \hat{v}_p) \quad (5)$$

While Hughes et al. approximate density analytically using face distance [1], our system utilizes a PTZ camera to directly maximize resolution. We calculate the effective area A_f by summing the geometric alignment score for every pixel p that currently falls on the face:

Where $\mathcal{P}_f$ is the set of all camera pixels observing face f in the current frame, $\hat{n}_f$ is the surface normal, and $\hat{v}_p$ is the viewing vector for that specific pixel. As the camera zooms in, the cardinality of the set $|\mathcal{P}_f|$ increases, naturally increasing the reward A_f .

VII. METHODOLOGY

A. Neural Network Architecture

The RGB and Semantic 2D image output from the Inspection and Navigation camera is passed through a 2D feature extractor, while the current local map of the occupancy, visibility, and visitation gets fed to a 3D feature extractor (Map encoder), and finally, both feature outputs are concatenated

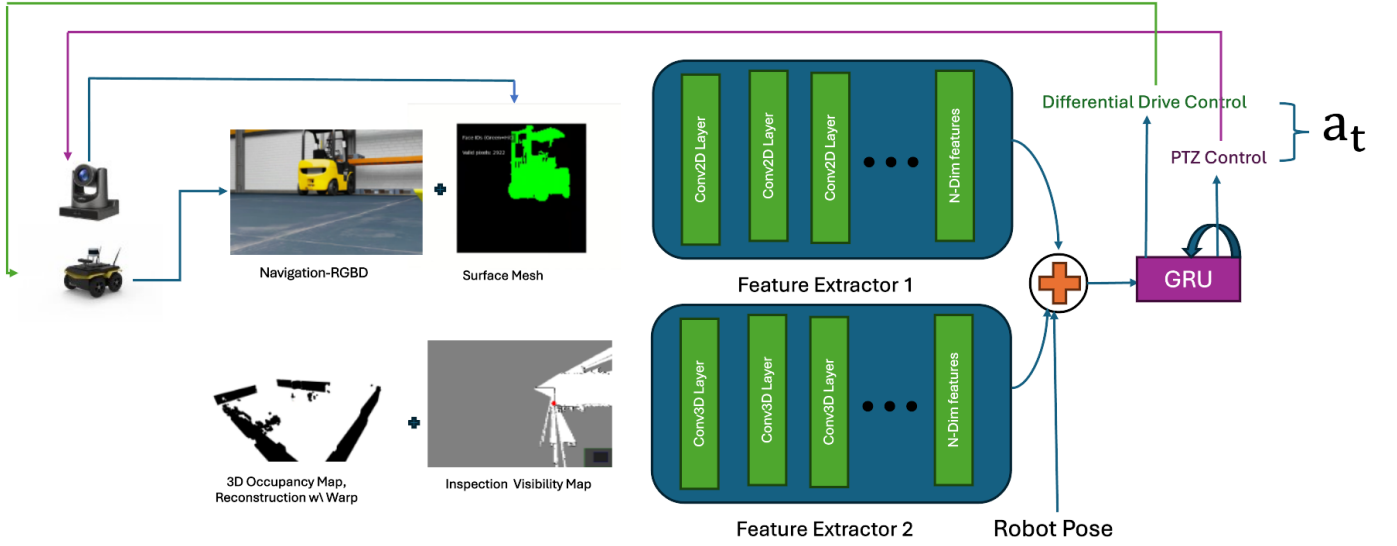


Fig. 1. System Architecture

with N-D feature vectors robot state S_t that encode the position and orientation of the robot. The outputs are fed into a 32-sequence GRU (Gated Recurrent Unit) network to encode temporal information, which is then passed to the Policy and Value MLP networks.

B. Ego Centric Local Map

To provide our agent with spatial information of the current state, we provide a local ego-centric map of dimension

$$[n, n, \frac{n}{2}, c]$$

where $n = 2\text{meters}$ and c represents the 3 channels described in section VI-A

C. Learning Joint policy for Inspection and Navigation

1) *Reward Design*: The overall reward obtained at each time step t is provided below

$$R(s_t, a_t) = R_{\text{inspect}} + R_{\text{exploration}} + R_{\text{penalty}}$$

2) *Inspection Reward R_{inspect}* : We compute the inspection reward R_{inspect} using a submodular marginal gain function in the inspection quality (VI-E) across all faces observed [1]

$$F_t = \alpha \sum_f \max(0, q(A_{f,t}) - q(A_f^{\text{best}}))$$

We note that a dedicated camera is used for Face ID detection in the simulator and would only be available during training.

3) *Exploration Reward $R_{\text{exploration}}$* : Our exploration reward optimizes the joint objective function $J(\pi)$ in equation 4.

$$R_{\text{exploration}} = \mathcal{I}_t(\mathcal{M}) + \mathcal{V}_t \quad (6)$$

$\mathcal{I}_t(\mathcal{M})$ represents the information gain of our occupancy map using the Shannon entropy. We define the information gain at time step t as

$$\mathcal{I}_t(\mathcal{M}) = \gamma_1(\mathcal{H}(\mathcal{M}_{t-1}) - \mathcal{H}(\mathcal{M}_t))$$

Where the total map entropy $\mathcal{H}(\mathcal{M}_t)$ is defined in equation 2

$$\mathcal{H}(\mathcal{M}_t) = \sum_{i=1}^N \mathcal{H}_i(\ell_{i,t}) \quad (7)$$

$\mathcal{V}_t$ represents the Visibility Increase Reward to encourage viewing as many Surfaces as possible. We reward viewing Surfaces, with the reward diminishing as we view the same surface.

Finally, we can define the full visibility increase reward $\mathcal{V}_t$ as the information gain over the surface visibility map.

$$\mathcal{V}_t = \mathcal{S}(\mathcal{V}_t) - \mathcal{S}(\mathcal{V}_{t-1})$$

4) *Camera Action Penalty*: To discourage jitter behavior of our PTZ Camera, we added a penalty for large changes in the control value. We compute this penalty with

$$\Delta u_t^{\text{ptz}} = \sum_{i \in (\text{pan, tilt, zoom})} (u_{t,i} - u_{t-1,i})^2$$

VIII. IMPLEMENTATION

Training is performed with a slight modification to the PPO-GRU (Proximal Policy Optimization [2]) implementation provided by [3], SKRL: Reinforcement Learning Library. To encode the 3D information, including the local occupancy and visibility maps, as well as the 2D RGB and semantic Images from the Inspection and Navigation camera, we use the 3D/2D ResNet encoder model from [4].

IX. TRAINING ENVIRONMENT AND SIMULATION

Trained on a NVIDIA RTX 8000 using Isaac Lab [5]. We use a modified Jackal by creating a custom PTZ articulation, increasing the control dimensions to 7 4 for the wheels, 2 for Panning and tilting, and 1 for zooming. The Navigation Camera and Inspection Camera (PTZ) are implemented with the Tiled Camera and the Ray Caster Camera. All Cameras are RGBD Cameras. The Ray Caster camera provides us with special information about the discretized faces of a mesh.

X. EXPERIMENTATION AND ABLATION STUDIES

A. Experimental Setup

To validate our method, we use the open-room Warehouse environment (with forklifts) from the publicly available Isaac Lab Assets.

We add additional constraints to the environment using the basic Prims available (Cone, Sphere, etc.). With randomized positions and properties. Using a 2D grid, we have a fixed set of evenly distributed safe positions, and we randomly initialized the positions of both the robot and the Inspection target. For generalizable capabilities, we also randomly initialize diverse inspection objects to be inspected.

Here, I write a list for myself of things and comparisons to make

- Frontier Exploration
- Frontier Exploration + RL for PTZ camera
- Random sampling methods with view-based planning
- Compare performance without any maps
- Compare performance without visibility maps
- Compare performance without Occupancy maps
- Compare performance without Semantics maps
- Compare performance with fixed Camera

XI. REAL-WORLD DEPLOYMENT

XII. FUTURE WORK

Our future work involves deploying our agent on a Real-World Jackal robot and Pan Tilt Zoom Camera. Towards the goal of building a more generalist agent we aim to prompt the agent with Natural Language

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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